

Knowledge-Driven Architecture for Alzheimer's Disease Research

A Multi-Modal Data Integration & Causal Discovery Framework

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Research Question

This study investigates whether the causal chains governing amyloid accumulation, tau propagation, and neurodegeneration in Alzheimer's disease can be temporally characterized through the integration of knowledge graphs derived from static ADNI clinical data with longitudinal functional and structural connectivity measures obtained via fMRI and DTI.

1. Introduction and Motivation

Alzheimer's Disease (AD) research generates vast multi-modal datasets spanning neuroimaging, cerebrospinal fluid biomarkers, cognitive assessments, genetics, and clinical observations. The Alzheimer's Disease Neuroimaging Initiative (ADNI) has produced one of the richest longitudinal datasets available, encompassing over 180 tables with more than 5,800 data columns across multiple study phases (ADNI-1, ADNI-GO, ADNI-2, ADNI-3, ADNI-4). Despite its comprehensiveness, the ADNI dataset remains fragmented: clinical CSV tables and DICOM imaging data exist in separate silos with no unified query interface that preserves semantic context.

This project constructs a semantically enriched, causality-aware Knowledge Graph (KG) that integrates the full breadth of ADNI data into a single queryable structure. Moving beyond simple data warehousing, the system aims to enable mechanistic understanding of disease progression—specifically the amyloid–tau–neurodegeneration (A→T→N) cascade—through formal ontology grounding and causal discovery algorithms. The central research question investigates whether causal chains governing amyloid accumulation, tau propagation, and neurodegeneration can be temporally characterized by integrating knowledge graphs derived from static clinical data with longitudinal functional and structural connectivity measures from fMRI and DTI.

2. System Architecture and Current Implementation

The system employs a multi-tier architecture consisting of four integrated components: a Neo4j-based Labeled Property Graph (LPG), Elasticsearch for metadata indexing, Redis for query caching, and a multi-tier file system for lossless image storage. A 15-step automated ETL pipeline ingests raw ADNI data—both clinical CSV tables and DICOM neuroimaging files—through sequential processing stages: database setup, table loading, patient entity creation, family history extraction, DICOM image processing with multi-tier storage (original DICOM, lossless 16-bit PNG/TIFF derivatives, JPEG thumbnails), clinical findings extraction, batch insertion, relationship creation, knowledge graph enhancement, query execution, biomarker analysis, graph enhancement, exploratory data analysis, and metrics computation.

Current Database Statistics

Entity	Count	Description
Patients	2,638	Unique ADNI participants across all phases
Visits	30,267	Clinical visits with temporal ordering (FOLLOWED_BY chains)
Diagnoses	25,946	Diagnostic assessments (CN, MCI, AD) with progression tracking
Total Nodes	~407,000	All entity types: patients, visits, images, biomarkers, genetics
Total Relationships	~1,160,000	Edges connecting entities across all data modalities

The patient-centric design places individual patients as hub nodes connected to visits, diagnoses, biomarkers, imaging data, medications, adverse events, family history, and genetic risk profiles. Each patient averages ~11.5 visits, with temporal chains via FOLLOWED_BY relationships and delta_months properties. Disease progression is captured through PROGRESSED_TO edges, and conversion events (CN→MCI,

MCI→AD) are represented as explicit ConversionEvent nodes. The pipeline achieves ingestion rates exceeding 500 images/second with sub-second retrieval, validated through a peer-reviewed publication at IEEE International Conference on Big Data 2025.

3. From Labeled Property Graph to Semantic Knowledge Graph

A critical distinction in this work is the difference between a Labeled Property Graph and a true Knowledge Graph. The current Neo4j implementation is an LPG: relationships are typed strings (HAS_DIAGNOSIS, TAKES_MEDICATION) without formal semantic grounding. A true KG requires ontology-grounded predicates—relationships that carry machine-interpretable meaning enabling reasoning and inference. The planned semantic enhancement transforms every entity and relationship to reference standard biomedical ontologies.

LPG vs. Knowledge Graph Comparison

Aspect	LPG (Current State)	True KG (Target State)
Schema	Implicit, application-defined	Formal OWL/RDF ontology
Relationships	Typed strings (syntactic)	Ontology-grounded (semantic)
Inference	Not supported	Reasoning capabilities
Interoperability	System-specific	SNOMED-CT, LOINC, UBERON, HPO
Query Language	Cypher only	SPARQL + Cypher

The target KG architecture is organized into eight layers: (1) Ontology Layer referencing SNOMED-CT, HPO, LOINC, UBERON, RxNorm, MONDO, Gene Ontology; (2) Patient Semantic Entity (foaf:Person, ncit:C16960); (3) Temporal Layer (OWL-Time); (4) Observations grounded in OBOE/LOINC; (5) Findings mapped to HPO; (6) Pathological States (AmyloidPathology, TauPathology, Neurodegeneration) linked by ro:RO_0002411 “causally upstream of”; (7) Disease entities from ICD/MONDO with conversion events; and (8) Treatments linked to RxNorm. All relationships use Relation Ontology predicates.

4. Temporal Graph, ATN Classification, and Biomarker Cascade

The temporal graph component serves a focused purpose: tracking patient sickness progression through longitudinal clinical journeys. FOLLOWED_BY relationship chains establish temporal ordering between visits, while delta_months properties capture the intervals between observations. This infrastructure enables queries such as “What was this patient’s cognitive trajectory over 5 years?” and “When did biomarker abnormalities first appear relative to clinical symptoms?”

The system implements ATN (Amyloid–Tau–Neurodegeneration) biological staging based on the NIA-AA Research Framework. Biomarker inputs—CSF analysis (Aβ42, Tau, pTau), Amyloid PET Centiloids, Tau PET SUVR, and MRI hippocampal volumetrics—feed into a classification decision tree producing four outcome profiles: A–T–N– (Normal), A+T–N– (Preclinical AD), A+T+N– (Early AD), and A+T+N+ (AD Dementia). This classification enables biological staging independent of clinical diagnosis, a critical capability for understanding the mechanistic cascade from amyloid deposition through tau propagation to neurodegeneration.

Importantly, the temporal component is scoped to patient trajectory tracking for the Master’s thesis, with dynamic temporal causal modeling—such as biomarker cascade simulation and personalized trajectory prediction—designated as future PhD-level research.

5. Causal Discovery Framework

The planned causal discovery pipeline consists of four stages: (1) Data Preparation—extracting variables from the KG with preprocessing for missing data and normalization; (2) Causal Discovery—running multiple algorithms in parallel: PC (constraint-based), FCI (handles latent confounders, recommended for medical data), GES (score-based), and DAG-GNN (neural network); (3) Integration—comparing results across algorithms, validating against literature, and embedding confirmed causal relationships as native KG edges (CAUSES, PREVENTS, PRECEDES); (4) Causal Inference—using the DoWhy framework for effect estimation and refutation tests to quantify effects such as “What is the causal effect of amyloid positivity on cognitive decline rate?” The architecture incorporates explicit confounder nodes (demographics, APOE genotype), temporal ordering through PRECEDES relationships, and integration with AlzKB (118,902 entities, 1,309,527 relationships) for prior knowledge constraints.

6. Scope and Roadmap

The thesis follows a three-phase roadmap within clearly defined Master’s-level scope. Phase 1 (Construct) builds the data-driven Knowledge Graph with ontology integration—this phase is substantially complete. Phase 2 (Discover) develops and compares causal discovery algorithms on the integrated data. Phase 3 (Represent) defines experiment protocols and embeds validated causal relationships into the graph. Future PhD-level work encompasses dynamic temporal modeling, personalized trajectory prediction, and clinical validation.

Master’s Scope (Current)	Future PhD Scope
Construct → Discover → Represent	Predict → Personalize → Validate
Static KG + causal discovery algorithms (PC, FCI, GES)	Dynamic temporal causal graphs
Ontology grounding (SNOMED-CT, LOINC, UBERON)	Biomarker cascade simulation (A→T→N)
Baseline-only causal analysis	Personalized trajectory prediction
Literature-validated causal overlay	Clinical trial emulation & validation

The thesis defense is scheduled for May 2026. The project is supported by an existing collaboration between Galatasaray University and CY Cergy Paris University (ETIS Lab), with ongoing consortium building for the EP BrainHealth 2026 Joint Transnational Call targeting Call 2: Neurodegeneration (application deadline: 10 March 2026). The open-source codebase is available at github.com/theFellandes/ADNIKnowledgeGraph.

7. Publications

- **Published:** Güngör, O., Turhan, S. N., Pınarer, Ö., Arib, S. “A Knowledge-Driven Multi-Tier Architecture for Lossless Medical Image Storage and Sub-Second Retrieval in Large-Scale Neuroimaging Studies.” *IEEE International Conference on Big Data 2025*.
- **In Progress:** Systematic literature survey on Knowledge Graphs, causal modeling, and data integration methods for AD research.
- **In Progress:** Graph modeling and causal overlay—semantic enrichment of the LPG with ontology grounding and causal discovery integration.